MRI radiomics and deep learning for muscle-invasive bladder cancer: how much of the reported accuracy survives leakage-free validation?

Public four-centre T2WI dataset

275 tumours, 221 patients

85 MIBC (31%)

Pre-specified models:

radiomics, fine-tuned CNN, deep-radiomics fusion

Comparator: tumour size

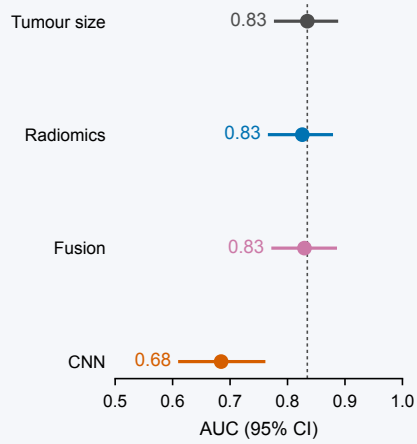
Leakage-free validation:

patient-grouped CV and held-out centres (primary)

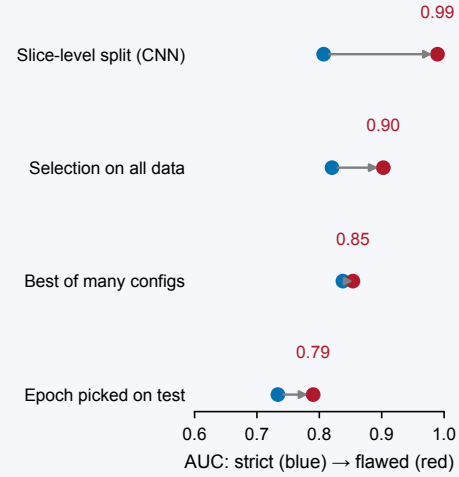
Leakage audit:

one evaluation flaw at a time

Held-out centres: no gain over size



One flaw at a time inflates AUC



Without evaluation leakage, T2WI radiomics and deep learning did not detect muscle invasion better than tumour size; single evaluation flaws inflated AUCs up to the near-perfect values reported in the literature.